

MOHAMMED SAEED MOHAMMED (PhD, P.E., P.Eng.)

Reno NV, USA • mosaeedeng@gmail.com • 775.200.3973 • www.linkedin.com/in/mohammedsmohammed

~ ENGINEERING DIVISION MANAGER ~

Polished, accomplished, and results-driven professional with extensive experience in structural and earthquake engineering, project management and execution, research and development, product innovation, business development, business processes, and team building. Skilled in leading organizations and teams to new levels of performance in domestic and international industries. Proven track record of improving efficiency, operational stability, and building diverse engineering teams in different countries that manage complex projects, and effectively deliver solutions to achieve performance goals and targets. An industry expert in seismic design and rehabilitation of buildings and bridges in regions of high seismicity. Skilled in utilizing emerging and innovative technologies such as seismic isolation, energy dissipation, and re-centering systems to maximize value and improve resilience of our cities and communities. The proficiency in project management, coupled with the technical acumen, has been pivotal in successfully completing high-impact projects across various global regions such as USA, Canada, Mexico, New Zealand, Peru, Chile, Colombia, Turkey, Japan, South Korea, Azerbaijan, Egypt, Guatemala, Argentina, Nicaragua, and India. An active member of the code development committees for ASCE 7, ASCE 41, and ASCE 4. Selected among ENR Southwest's top young professionals.

Proven expertise in:

- | | | |
|---|--|---|
| • Structural Engineering | • Seismic Protection Technologies | • Research & Development |
| • Project Management & Execution | • Data Analysis | • Business Development |
| • Risk Management and Mitigation | • Strategic Planning | • Resource Allocation and Budget Management |
| • Code Development | • Leadership & Team Building | • Compliance with Building Codes and Safety Regulations |
| • Seismic Design & Rehabilitation of Structures | • Effective Stakeholder Communication and Engagement | |

EDUCATION

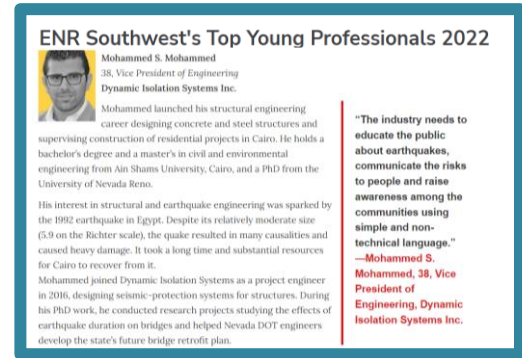
- | | |
|---|------------|
| Imperial College Business School | 2025 |
| • Master of Business Administration (MBA) | (expected) |
| University of Nevada, Reno | 2016 |
| • Doctor of Philosophy (Ph.D.), Civil and Environmental Engineering | |
| Ain Shams University, Cairo, Egypt | 2011 |
| • Master of Science (M.Sc.), Civil Engineering "Structural Engineering" | |
| Ain Shams University, Cairo, Egypt | 2005 |
| • Bachelor of Science (B.Sc.), Civil Engineering "Structural Engineering" | |

PROFESSIONAL REGISTRATIONS, AWARDS, & AFFILIATIONS

- Selected Among ENR Southwest's Top Young Professionals for 2022
- Registered Professional Engineer, NV (P.E. License No. 028445)

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- Registered Professional Engineer, Saskatchewan (P.Eng. License No.73782)
- Registered Civil Engineer, Egyptian Engineers Syndicate (License No. 40-00865)
- Chartered Engineer (UK) (in progress – 2024)
- American Society of Civil Engineers (ASCE), member
- Earthquake Engineering Research Institute (EERI), member
- Structural Engineers Association of Northern California (SEAONC), member



EXPERIENCE

OSF ENGINEERS (Reno NV, USA)

January 2024 (under establishment)

Principal / Engineering Division Manager

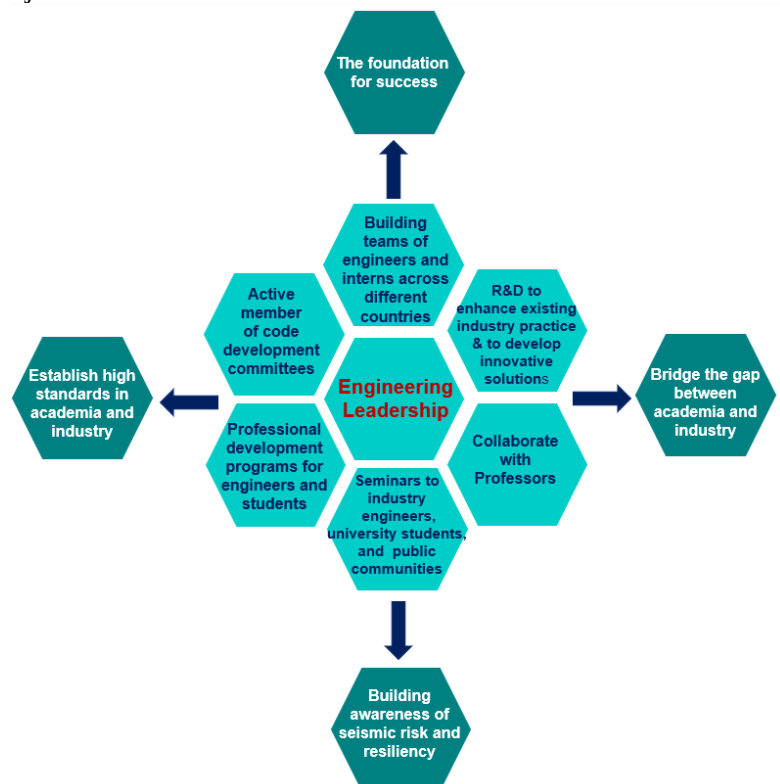
DYNAMIC ISOLATION SYSTEMS (DIS) (Reno NV, USA)

August 2016 to December 2023

Vice President/Director of Engineering:

Build and manage multidisciplinary cross-functional engineering teams across different countries. The engineering teams focus on analysis, design, development, testing, and project management of seismic protection systems for different structures in seismic countries. Work closely with the organizational leadership and with Kawakin Core-Tech; DIS' parent company in Japan, to grow and expand internationally. Mohammed uses his experience and education to lead the engineering and project management teams to develop innovative solutions and successfully deliver major earthquake protection projects.

- Develop and implement strategic plans for the engineering department
- Deliver industry leading seismic protection solutions and provide technical guidance to the engineering teams
- Work with the engineering team in the US on new business and technology development for emerging applications such as data centers and dams
- Lead and mentor project managers to ensure the successful execution of diverse construction projects by fostering a culture of excellence, efficiency, and innovation in project management practices
- Research and development of new products that provide seismic protection for different structural and non-structural applications
- Designed seismic protection systems for more than 100 projects including bridges (new and retrofit), hospitals, commercial buildings, silos, convention centres, tall



buildings, dam control equipment, artwork, data centers, condominiums buildings, airports, temples, and wharves

- Contract reviews and negotiations
- Establish KPIs and reporting processes to measure and improve efficiencies among different teams
- Collaborate with other departments within the organization on improving company's processes, addressing issues, and establishing short, medium, and long-term strategies to grow the company
- Emphasized and pushed for industry- academia collaboration through several projects with the University of Nevada Reno, the University at Buffalo, University of California Berkeley, University of British Columbia, University of Virginia, Ankara Yıldırım Beyazıt University, and others
- Lead the engineering team on conducting several shake table tests at the University of Nevada Reno and the University of California San Diego to examine and validate innovative solutions that were used to protect structural and non-structural systems against extreme seismic events
- Increased the overall sales by more than 100% through implementing new and innovative design techniques and optimization
- Launched an internal program of professional development courses, webinars, and presentations for the engineering teams
- Prepared a program to educate industry engineers, university students, and the public in earthquake prone countries about seismic risk, resiliency, sustainability, and seismic protection technologies. Gave presentations to public organizations, universities, and engineering firms (like WSP, HDR, TY Lin, and others)
- Improved the awareness among the engineers and the public about the importance of resilience for our cities and communities
- Improved productivity by 35% and efficiency by 20% in 12 months by improving design methods, and implementing design optimization techniques
- An active member of the code development committees for ASCE 7, ASCE 41, and ASCE 4
- Responsible for all the engineering and technical activities in the company:
 - Review and approve design submittals and test reports associated with assigned projects
 - Review contracts and technical specifications
 - Review and approve design and details of seismic isolation and damping systems for structures
 - Structural modeling and analysis using SAP2000 and ETABS
 - Evaluation of product test data
 - Preparation of technical and commercial Proposals
 - Production engineering support
 - Product development of dampers, friction pendulums, and equipment isolation systems
 - Research and development of new products that provide seismic protection for structures

UNIVERSITY Of NEVADA RENO (UNR) (Reno NV, USA) *August 2012 to December 2016*

Research & Teaching Assistant:

Worked on two projects:

- 1) ***Effect of ground motion duration on bridge columns, Sponsor: FHWA*** 2013-2016
 - Successfully tested 5 RC bridge columns on shake tables using long & short duration ground motions, demonstrating precise time management and project scheduling to meet key milestones.
 - Collaboratively worked with contractors and laboratory personnel in the design, construction, and instrumentation of the specimens, showcasing effective budget management and resource allocation skills to ensure project efficiency and cost-effectiveness.

2) ***Path from earthquake to bridge evaluation: Implementation of ShakeCast in Nevada,*** 2012-2015
Sponsor: Nevada Department of Transportation (NDOT)

- Successfully implemented ShakeCast with the Nevada Bridge Inventory and collaborated with the Nevada Seismological Laboratory to develop over 100 earthquake scenarios. This involved evaluating the impact of these scenarios on Nevada's bridges and compiling a critical list of the most vulnerable bridges. These efforts were instrumental in informing the future state bridge retrofit plan, showcasing my capabilities in project execution, analytical assessment, and strategic infrastructure planning

CONSULTING ENGINEERING OFFICE (CEO) (Cairo, Egypt) ***January 2006 to July 2012***

Project Structural Engineer:

- Assisted in proposing various structural systems and developing computer models under the guidance of senior engineers, contributing to project planning and execution
- Supported the structural and seismic analysis and design of steel and reinforced concrete structures, collaborating with senior engineers to manage project phases and interdisciplinary coordination
- Participated in the supervision of residential project construction, aiding in overseeing project schedules, resource allocation, and quality assurance
- Helped prepare tender and construction packages, including shop drawings, specifications, and bill of quantities, under senior engineers' supervision, focusing on project documentation and compliance
- Contributed to checking and revising structural designs, preparing technical reports, and participating in project review meetings to ensure consistency and adequacy of designs
- Involved in project communication and coordination, facilitating smooth workflow and issue resolution under the guidance of senior project managers
- Supported project budget management, including cost estimation and budget tracking, and contributed to financial reporting.
- Aided in negotiations with suppliers and vendors, ensuring quality and cost-effectiveness in procurement under senior management direction
- Coordinated with regulatory bodies for project approvals and compliance, under the mentorship of senior engineers
- Supported strategic project planning and decision-making processes, focusing on resource allocation and project prioritization
- Assisted in maintaining comprehensive project documentation for transparent communication with stakeholders
- Selected projects:
 - Grain storage silos and catwalks in Giza, Minya and Qena, Egypt
 - 85 Residential buildings-Al Azhar University Faculty residential buildings, 5th Settlement, Cairo, Egypt
 - Shopping center, Ismailia, Egypt
 - International youth hostels in Sharm El-Sheikh and Marsa Matrouh, Egypt
 - Column retrofitting of a 15-story building in Alexandria, Egypt

AIN SHAMS UNIVERSITY (Cairo, Egypt) ***May 2006 to July 2012***

Teaching Assistant:

CES 346: Design of Steel Structures & CES 446: Design of Steel Bridges

- Participation in curriculum building and development, and implementation of creative and innovative teaching strategies
- Holding discussion & laboratory sessions for 150+ undergraduate students (10 hours per week)
- Participation in preparing lecture notes & homework assignments, grading quizzes and mid-term exams

Representative Project Experience

1- UNR Soil Box

- Location: Nevada, USA
- An innovative large-scale laminar Soil Box system was completed as part of the earthquake engineering complex at the University of Nevada Reno. The Soil Box system is the largest in the US and comparable in size to the largest in the world. It is 24-foot square, 25-foot tall integrated shake table and soil box that has the capacity to hold 350 tons of soil, plus a structure on top. The Soil Box system provides an innovative facility to evaluate experimentally soil-structure interaction during an earthquake. Previous systems used mechanical bearings across the heights of the boxes, however, several failures happened during testing. The designers for the UNR soil box decided to use elastomeric/rubber bearings instead of the mechanical ones. The soil box has 19 layers, called laminates, that are each supported on rubber bearings so that the soil layers can move relative to each other like soil does during actual earthquake. Mohammed worked with the team from UNR on the design of the elastomeric bearings. Extensive research was conducted to evaluate the bearings performance under extreme tension loading. (Photo sources: 2news.com and UNR)

<https://www.unr.edu/nevada-today/news/2022/soil-box-design-team>



2- Nicaragua Bridges

- Location: Nicaragua
- The bridges shown in the photos are considered the first seismically isolated projects in Nicaragua. Mohammed educated the design engineers on seismic resilience and how to protect their structures against earthquakes. Mohammed also assisted with the design and approvals with the local government. (Photo source: DIS)



3- Oregon Treasury Resiliency Building

- Location: Salem, Oregon, USA
- 35,000 ft² – 2 story – steel moment frames
- As part of its emergency planning work, several years ago, Oregon State Treasury identified a need for a resilient workplace. Mohammed worked with KPFF Engineers and GBD Architects on the design and testing of the seismic isolation system for the Oregon Treasury Resiliency Building. It is the first USRC platinum building in Oregon and the first USRC-rated seismically isolated building in the United States. (Photos source: The New York Times)

<https://www.nytimes.com/2022/04/08/us/oregon-state-treasury-building.html>

<https://www.usrc.org/portfolio-items/treasury-resilience-building-salem-or/>



4- The Harbor-171 retrofit

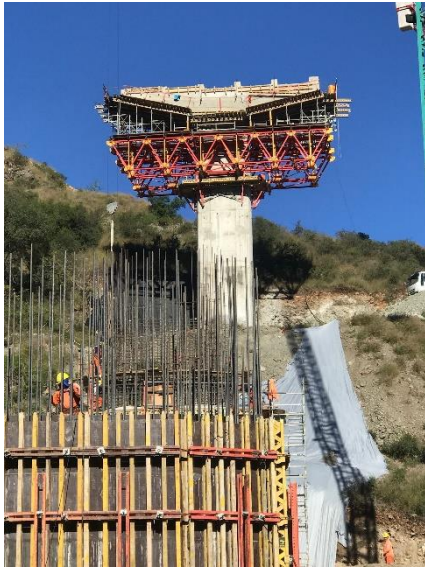
- Location: Puerto Vallarta, Mexico
- A 14-story residential building that was built in early 2000's. Because of financial reasons, only the skeleton was built and the building was not completed/occupied. The lateral resisting system was concrete moment frames. In 2017, the owner decided to continue with the construction and occupy the building. However, the seismic hazard at the location of the building has increased and the story drifts exceeded the code limits. Mohammed participated in meetings with the EOR and conducted feasibility study to use viscous wall dampers (VWDs). Mohammed recommended to retrofit using VWDs along the height of the building to absorb the earthquake energy and reduce the drifts. Mohammed generated ground motion acceleration histories that the EOR used to conduct nonlinear-history analysis and study the response of the building with and without the dampers. Mohammed designed the walls and the connections with the existing beams. Mohammed reviewed the strengthening scheme that the EOR designed for the bays where the wall-dampers were installed. Mohammed provided technical support to the contractor while installing the dampers within the building. (Photos source: DIS)



5- Viaductos Ruta Provincial N34

- Location: Cordoba, Argentina
- The viaduct is a new, six-span, prestressed-concrete bridge. Mohammed provided technical support for the EOR and performed ELF analysis for the seismically isolated bridge. Mohammed prepared a design report and shop drawings for the bearings and the

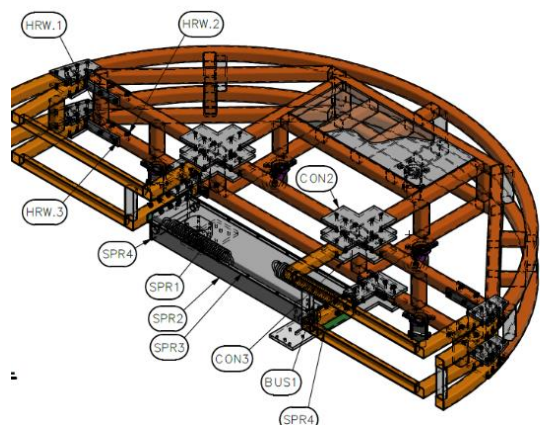
connections. Mohammed was in charge of the bearing testing and prepared test reports. Mohammed provided technical support for the contractor and also performed a general inspection during construction to ensure compliance with the design, project drawings and installation manuals.



6- Party Hat

- Location: San Diego, CA, USA
- Mohammed and the engineering team designed an innovative seismic isolation system to protect an extremely valuable piece of artwork against earthquakes. The Party Hat is placed in the main lobby of Jacobs Medical Center at UC San Diego. (Photo by Philipp Scholz Rittermann)

<https://today.ucsd.edu/story/monumental-sculpture-by-jeff-koons-debuts-at-jacobs-medical-center>



7- Representative Projects with US DOT's

a) Caltrans

➤ Scofield Avenue Undercrossing

The Scofield Avenue Undercrossing is an eleven-span steel girder bridge supported on skewed steel outrigger bents that was built in 1956. The bridge passes over a Chevron refinery which made the retrofit complicated. Mohammed provided technical support before the project started. Mohammed participated in meetings with Caltrans and reviewed the construction sequence and bearing-replacement procedure. Mohammed prepared a full design report that included the design of the bearings and shop drawings. The report was reviewed and stamped by a California PE. The prototype testing of the isolation bearings was conducted at UCSD. Mohammed prepared the testing protocols and was in charge of the testing. Mohammed prepared test reports that were then approved by Caltrans. Mohammed provided technical support to the contractor during construction.

b) WYDOT

- Mohammed worked with WYDOT and different EOR's on retrofitting several bridges crossing the Snake River

8- Representative Projects in New Zealand

a) Wellington Convention and Exhibition Centre



(Photo source: WellingtonNZ)

b) Kumutoto Site 10 office building



(Photo source: athfield architects)

c) Bob Scott retirement Village

Mohammed designed seismic base isolators that protected the village against the Mw 7.8 Kaikoura earthquake in November 2016 (Photo source: Ryman Healthcare)

<https://student-learning.in/2021/10/04/4-pillars-of-building-academic-culture-passionate-purposeful-and-participative-campus/>

